Porting Lava autonomous-QA solutions â†’ Boba-Base (submodules-driven)

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Maps the Lava deep-research playbook (`Lava:docs/autonomous-qa/PORTING-PLAYBOOK.md`) onto **Boba-Baseâ€™s actual stack** (Go `qBitTorrent-go` + Python `download-proxy/merge_service` + `plugins` + `frontend/extension` + Jackett), respecting Boba-Baseâ€™s **TDD + anti-bluff (CONST-XII)** mandate. Both repos already share `submodules/{containers, jackett, challenges, helixqa}` + `constitution`, so the reusable pieces land **in the shared submodule** (consumed by both), and only thin glue stays per-repo.

Applicability triage (honest â€” not a blind copy)

Lava Â§	Solution	Applies to Boba-Base?	Why
Â§1	Containerized **Android** emulator	â�Œ no	Boba-Base has no Android; UI is web (`frontend/extension`) â†’ browser E2E (Playwright-in-container) is the analog, separate work
Â§2	Android guest-net `ndc` fix	â�Œ no	Android-only
Â§7	Compose UI Challenge + off-main nav	â�Œ no	Android-only
Â§0/Â§4	**Egress-path insight + VPN-host egress**	âœ… **YES â€” primary**	Same datacenter-IP tracker block; affects Jackett + `merge_service` + `download-proxy` + plugin engines
Â§3	**Configurable outbound proxy**	âœ… **YES**	`download-proxy`/`qBitTorrent-go`/Jackett tracker calls need a configurable egress
Â§5	**Durable remote execution (systemd-linger)**	âœ… **YES**	`run_all_challenges.sh`/`ci.sh`/`helixqa` long runs on remote hosts get reaped by logind

Lava Â§	Solution	Applies to Boba-Base?	Why
Â§8	**Jackett cookie-login + cookie-harvest**	âš ï¸� partial â€” already present	Boba-Base has nnmclub-cookie-*, extract-tracker-cookies.sh, extract-jackett-key.py; ADD the apikey-vs-cookie management distinction + test fake-equivalence
Â§6/Â§9	**Anti-bluff QA verdict + diagnostics**	âœ… YES	strengthen run_all_challenges.sh verdicts + the diagnostic gotchas

The four high-value ports (in order)

P1 â€” Durable remote execution (Â§5) [submodule: containers; glue: scripts/]

Problem Boba-Base shares: long QA/deploy runs on a remote host die when the SSH session ends. **Root cause:** remote systemd-logind KillUserProcesses reaps detached user processes (tmux/nohup/setsid all die). **Fix:** loginctl enable-linger + systemd-run --user --unit=<n> --collect bash runner.sh; poll systemctl --user is-active; sentinel file for completion. Avoid piping the long command through tail -N (buffers until exit) â€” log per-step artifacts. **Port:** add containers submodule helper pkg/remoteexec (or scripts/lib/durable-run.sh) shared by both repos; wire scripts/deploy-remote.sh + run_all_challenges.sh to use it. **TDD:** a test that launches a 60s sleeper via the helper, drops the SSH session, and asserts itâ€™s STILL running after (fails against the old nohup approach).

P2 â€” Egress decision + VPN-host routing (Â§0/Â§4) [submodule: containers; glue: scripts/ + Jackett/proxy env]

Problem Boba-Base shares: on a datacenter host, trackers are network-blocked (DNS-fail/TLS-MITM, **not** Cloudflare â†’ FlareSolverr canâ€™t fix). Affects Jackett indexer fetches + merge_service/download-proxy + plugin engines. **Diagnosis (port the script):** curl https://api.ipify.org (host IP) + curl -o /dev/null -w %{http_code} https://<tracker>/ direct vs via a VPN-host SOCKS proxy. Different egress IP + 200 via proxy â‡’ confirmed. **Fix:** route outbound through a VPN-connected host (the nezha pattern). SOCKS tunnel ssh -D 127.0.0.1:1080 -N <vpnhost> (use --socks5-hostname for remote DNS); point Jackett + download-proxy + qBitTorrent-go at it (P3). For browser-cookie harvest, run the harvester ON the VPN host. **Port:** containers submodule pkg/egress (tunnel up/verify) + scripts/egress-via-vpn.sh glue; reuse Boba-Baseâ€™s existing ensure-macos-tunnel.sh style. **TDD:** assert the via-proxy egress IP â‰ direct host IP AND a known-blocked tracker returns 200 via proxy.

P3 â€” Configurable outbound proxy in the services (Â§3)

download-proxy (Python): `httpx/requests` honor `HTTP_PROXY/HTTPS_PROXY/ALL_PROXY/NO_PROXY` env natively â€” add an explicit `BOBA_UPSTREAM_PROXY` config that sets these for tracker-bound clients, with loopback bypass (`NO_PROXY=127.0.0.1,localhost,jackett`). **qBitTorrent-go:** set `http.Transport.Proxy` (socks5 native, remote DNS) from a `BOBA_UPSTREAM_PROXY` env â€” mirror Lava `internal/httpx/proxy.go`. **Jackett:** has a built-in proxy setting (configure via its API/ServerConfig). **Deploy gotcha (port):** the env must be FORWARDED into the containers (`docker-compose.yml` env / the boba-ctl deploy) â€” Lavaâ€™s bug was a missing allow-list entry. **Verify on distroless via `podman inspect`, not `exec printenv`. TDD:** a test with a local proxy asserts the serviceâ€™s tracker request traverses it; falsifiability: disable the wiring â†’ test fails.

P4 â€” Jackett cookie-login hardening (Â§8) [submodule: jackett integration + helixqa fakes]

Add to Boba-Baseâ€™s existing cookie infra: Jackettâ€™s **management API** (`/api/v2.0/indexers`, indexer `/config`) needs a **dashboard session cookie** (empty-password `POST /UI/Dashboard`) â€” the apikey only authorizes Torznab `/results+/caps`. If `boba-jackett/extract-jackett-key.py` only uses the apikey, `ListIndexers/config` silently 302â†’`/UI/Login`. **Critical for anti-bluff:** the test fake MUST 302-without-cookie like real Jackett (behavioral equivalence) or the gap stays hidden. **TDD:** fake 302s management without the cookie; assert discovery succeeds via the cookie path; falsifiability: remove cookie login â†’ 302 failure.

Submodules-driven principle (per Decoupled Reusable Architecture)

- **`containers` submodule** is the home for P1 (durable-exec) + P2 (egress/tunnel) â€” generic, both repos + future projects consume. Contribute upstream first, then bump the pin in Lava AND Boba-Base.
- **`jackett` integration + `helixqa`** is the home for P4â€™s cookie-login + the behaviorally-equivalent fake.
- **Per-repo glue only:** P3â€™s env wiring (each serviceâ€™s HTTP client + each repoâ€™s compose/deploy).

Execution order (TDD per CONST-XII â€” each step REDâ†’GREEN with pasted evidence)

1. P1 durable-exec helper in `containers` + wire `run_all_challenges.sh/deploy-remote.sh`.
2. P2 egress diagnosis + VPN-host tunnel helper in `containers` + `scripts/egress-via-vpn.sh`.
3. P3 `BOBA_UPSTREAM_PROXY` in `download-proxy` + `qBitTorrent-go` + Jackett + compose env-forward.
4. P4 Jackett cookie-login + fake-equivalence test.

5. Re-run `run_all_challenges.sh` from the VPN-routed host â†’ real tracker searchâ†’download evidence.

   Source playbook (full root-cause detail + Lava code refs): `Lava:docs/autonomous-qa/PORTING-PLAYBOOK.md`. Each P-item there has the exact mechanism + the falsifiability rehearsal to replicate under Boba-Baseâ€™s CONST-XII.